**USER INTERFACE DESIGN – EXPERIMENT 10**
**DATA VISUALIZATION USING JAVASCRIPT (CHART.JS)**

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**Course:** USER INTERFACE DESIGN

## AIM

The aim of this experiment is to create data visualizations such as pie charts and bar graphs for an inventory management system using JavaScript.

## PROCEDURE

An HTML file was created to structure the webpage and include canvas elements for displaying the charts. The Chart.js library was integrated using a CDN link to enable easy creation of visualizations. Basic styling was applied using CSS to align the charts and improve the visual appearance of the page. A JavaScript file was then created to define the inventory data, including categories such as electronics, clothing, home appliances, books, and toys along with their respective stock values. Using Chart.js, a pie chart was generated to represent the distribution of inventory items across different categories, and a bar graph was created to display the quantity of items in stock for each category. The charts were configured with appropriate colors, labels, and responsive settings to ensure clarity and readability.

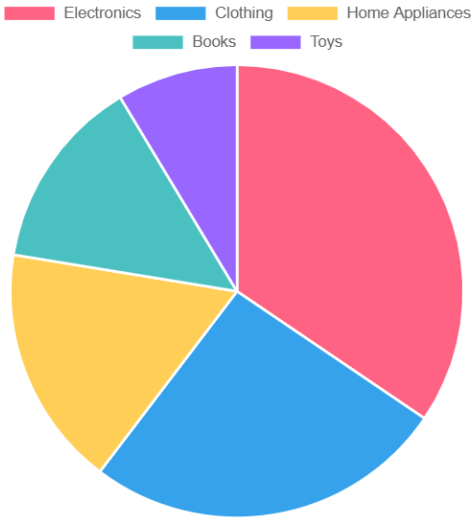
## OUTPUT DESCRIPTION

The output displays both a pie chart and a bar graph representing the inventory data. The pie chart illustrates the proportional distribution of items across various categories, allowing easy visualization of which category has the highest or lowest stock. The bar graph provides a clear comparison of the number of items available in each category, making it easier to analyze stock levels. Both charts are displayed clearly with proper labels and colors, enhancing user understanding of the data.

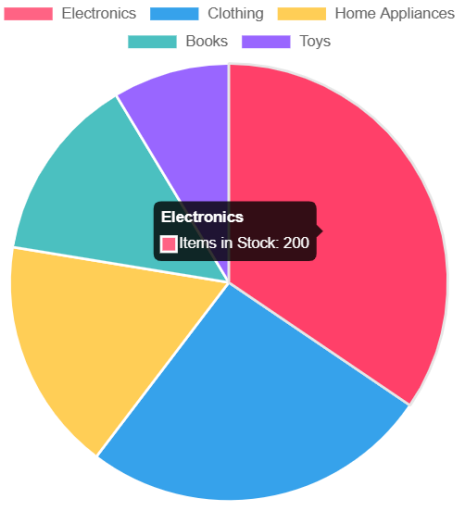
## RESULT

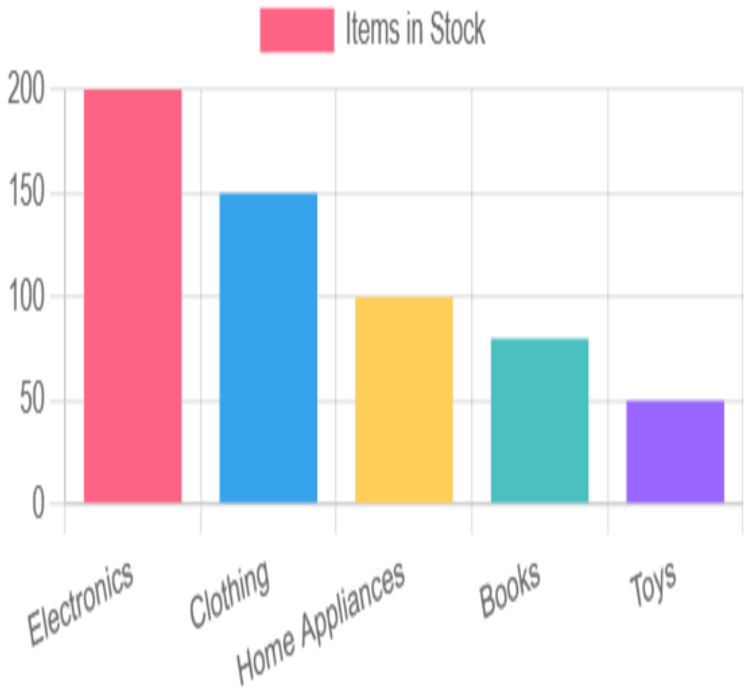
Thus, data visualization for an inventory management system was successfully implemented using JavaScript and Chart.js. The pie chart and bar graph effectively represent the inventory data, making it easier to analyze and interpret the information.

# Inventory Management System




# Inventory Management System






index.html.txt

File Edit View H1

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Inventory Management Visualization</title>

<style>
body {
    font-family: Arial, sans-serif;
    text-align: center;
    margin: 50px;
}

/* FIX: reduce chart size */
canvas {
    width: 400px !important;
    height: 400px !important;
    margin: 20px auto;
    display: block;
}
</style>

</head>
<body>

<h1>Inventory Management System</h1>

<canvas id="pieChart"></canvas>
<canvas id="barChart"></canvas>

<script src="https://cdn.jsdelivr.net/npm/chart.js"></script>
<script src="script.js"></script>

</body>
</html>
```

index.html.txt | script.js.txt

File Edit View

```javascript
// Inventory data
const inventoryData = {
    labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'],
    datasets: [{
        label: 'Items in Stock',
        data: [200, 150, 100, 80, 50],
        backgroundColor: [
            '#FF6384',
            '#36A2EB',
            '#FFCE56',
            '#4BC0C0',
            '#9966FF'
        ]
    }]
};

// PIE CHART
const ctxPie = document.getElementById('pieChart').getContext('2d');
new Chart(ctxPie, {
    type: 'pie',
    data: inventoryData
});

// BAR CHART
const ctxBar = document.getElementById('barChart').getContext('2d');
new Chart(ctxBar, {
    type: 'bar',
    data: inventoryData,
    options: {
        scales: {
            y: {
                beginAtZero: true
            }
        }
    }
});
```